Spine

SPINE Volume 44, Number 17, pp E992–E999
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BASIC SCIENCE

Transport of Vancomycin and Cefepime Into Human Intervertebral Discs

Quantitative Analyses

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Study Design. Simulation of antibiotics transport into human intervertebral disc with intravenous infusion.

Objective. The objective of this study was to quantitatively investigate antibiotic concentrations in the disc.

Summary of Background Data. Intravenous infusion of antibiotics is typically used to treat intervertebral disc infection in clinics. However, it is difficult to evaluate the drug concentrations within discs *in vivo*.

Methods. A computational model was used in this study. The variation of drug charge with pH was considered in the model. Thirty-minute infusions of two commonly used antibiotics in clinic—vancomycin and cefepime—were numerically investigated. Spatial and temporal concentration distributions of these drugs in both nondegenerated and moderately degenerated discs were calculated.

Results. For intravenous infusion of 1 g vancomycin and 2 g cefepime in 30 minutes repeated every 12 hours, it was predicted that vancomycin concentration in the disc fluctuated between 17.0 and 31.0 times of its minimum inhibitory concentration (1 ug/mL) and cefepime concentration fluctuated between 1.1 and 4.2 times of its minimum inhibitory concentration (*i.e.*, 8 ug/mL) in about 2 days. It was also found that vancomycin concentration in moderately degenerated disc was lower than that in the nondegenerated disc.

Conclusion. This study provides quantitative guidance on selecting proper dosage for treating disc infection. The method used in this study could be used to provide quantitative information on transport of other antibiotics and drugs in discs as well.

Key words: antibiotic penetration, cefepime, disc infection, finite element method, intervertebral disc, mixture theory, modeling, transport phenomena, vancomycin.

Level of Evidence: N/A

Spine 2019;44:E992–E999

Intervertebral disc (IVD) infection is a dangerous, devastating, and debilitating condition with growing incidence related to the aging population, the rise of intravenous (IV) drug abuse, an increase in spinal surgeries, diabetes mellitus, and hematogenous spread, particularly in immunosuppressed patients (*e.g.*, HIV, cancer, rheumatic disease patients).[1]

Treatment of disc infection often requires weeks of inpatient care and IV administered drugs, but it is difficult to evaluate the drug concentration inside discs. This is because the IVD is the largest avascular structure in human body, the transport of antibiotic drugs to the tissue mainly relies on diffusion. During disc degeneration or aging, discs become less hydrated, consequently, less permeable to drug and fluid transport.[2–5]

The diffusion coefficient, solubility, metabolic half-life, serum binding, molecular size of the drug could all affect the drug transport from blood stream to the disc. For example, an antibiotic with lower diffusion coefficient (usually with large size), less solubility, and short metabolic half-life may result in the poor antibiotic penetration into the disc, as the drug molecules may be metabolized before diffusing into the disc, leading to lower drug concentration within the disc.

The extracellular matrix of disc tissue is negatively charged, mainly due to the high content of glycosaminoglycans inside tissue,[6] making it easier for positively charged drugs and more difficult for negatively charged drugs to accumulate in the disc.[7–13]

Variations in fixed charge density, hydration, pH of the disc tissue also affect the antibiotics penetration into the disc. For example, the normal human disc has a pH value in the range of 7.0 to 7.4. Discs become more acidic (*e.g.*, pH

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Acknowledgment date: October 1, 2018. First revision date: November 29, 2018. Second revision date: January 31, 2019. Acceptance date: February 18, 2019.

The device(s)/drug(s) is/are FDA-approved or approved by corresponding national agency for this indication.

NIH/NIAMS (AR066240) and Miami Center for Orthopaedic Research and Education (Miami CORE) grant funds were received in support of this work.

Relevant financial activities outside the submitted work: grants, royalties.

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DOI: 10.1097/BRS.0000000000003028

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6.8) and less hydrated as the result of degeneration.[14,15] The charge of the antibiotic may vary with pH in disc, affecting its transport and concentration in the negatively charged disc tissue. In addition, the fixed charge density of disc tissue also changes with degeneration.[2–6,16]

The objective of this study was to numerically investigate antibiotic penetration into the IVD using a computational model. The results of this study could be used as guidance to develop new methods to treat disc infection and/or related disc degeneration. In this study, vancomycin and cefepime were used as examples of the antibiotics penetration into human discs for treating disc infections, as vancomycin and cefepime are commonly used in clinic for treating infections caused by a broad spectrum of bacterial species (*e.g.*, enterococci, staphylococci, *etc.*).[17]

METHODS

Finite Element Model

A previously developed finite element model[4,7] was extended to include pH-dependent charge of antibiotics (see below). Briefly, the disc geometry was taken from a human L2–3 disc (male, 41 yrs old, nondegenerated[18]), with a size of 27 mm anterior-posteriorly, 44 mm laterally, and 9 mm in height, see Figure 1A. The disc was modeled as an inhomogeneous structure with two distinctive regions: nucleus pulposus (NP) and annulus fibrosis (AF), Figure 1B. A part of vertebra was attached to the disc (Figure 1B). A quarter of the disc-vertebra was modeled due to symmetry (Figure 1B) with COMSOL software (COMSOL 5.3, COMSOL, Inc, Burlington, MA). The material properties of the disc-vertebra segment (including mechanical properties, transport properties, and electrochemical properties) for the nondegenerated disc and the moderately degenerated disc can be found in our previous study.[4,19] Briefly, for the nondegenerated disc, the distribution of the fixed charge density was: 0.3425 M in the NP, linearly decreased from 0.3425 to 0.1634 M from innermost to outermost AF; and the water content (volume fraction) distribution was: 0.85 in the NP, and linearly decreased from 0.85 to 0.7 from innermost to outermost AF. The moderately degenerated disc model (without annular tear) was virtually created by reducing the nutrition level at the boundary of the nondegenerated disc mentioned above. The details could be found in our previous study.[19] The disc at the 10th year of degeneration (after nutrition reduction) was collected and used as the moderately degenerated disc model in our present study. For this moderately degenerated disc, the viable cells were 40%, the fixed charge density was 80%, the disc height was 90.7%, of those of the nondegenerated disc, respectively.

Molecular Properties of Vancomycin Used in Our Model

Vancomycin is a glycopeptide antibiotic broadly used to treat infections caused by streptococci and staphylococci. The molecular weight of vancomycin is 1468.3 g/mol, the size of vancomycin is estimated with a mathematical relationship between molecular weight and size,[20] being around $r = 0.748$ nm. Diffusivity of vancomycin in water at 37°C calculated by the Wilke–Chang correlation is around 2.83×10^{-10} m^2/s.[21] Serum binding of vancomycin is around 55%.[22] That is, only 45% of vancomycin in serum is free, available for diffusion.

Molecular Properties of Cefepime Used in Our Model

Cefepime is a new parenteral fourth-generation cephalosporin with a broader spectrum of antimicrobial activity than most third-generation cephalosporin antibiotics. Cefepime demonstrates retaining potent activity against gram-positive organisms, Enterobacteriaceae, *Pseudomonas aeruginosa*. Cefepime molecular weight is 480.56 g/mol. Binding to plasma protein is 16%. The mean half-life of cefepime is approximately 2 hours and does not vary with dose or duration of dosing.[23] Diffusivity of cefepime in water at 37°C is 7×10^{-10} m^2/s.[24]

Change of Vancomycin Concentration in Serum With Time Following Infusion

Based on the measured serum vancomycin concentration in healthy subjects with normal renal functions by Krogstad

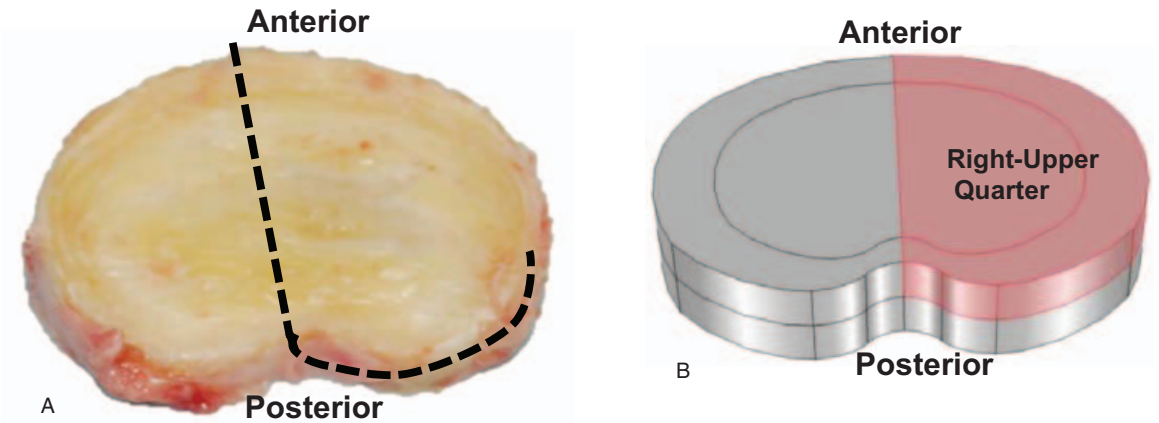

Figure 1. A, Picture of a human lumbar intervertebral disc (IVD, L2-3, nondegenerated). **B**, Schematic of a quarter of the disc used in the finite element model.

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et al,[22] the serum level of vancomycin after 1 g infusion in 30 minutes with a constant rate may be expressed as biexponential equation as follows:

$$C_{serum_van} = \frac{68.3}{0.5} \cdot t\,(t \le 0.5hr), \quad (1)$$

$$C_{serum_van} = 19.05.e^{-1.21\cdot(t-0.5)} + 49.25.e^{-0.05642\cdot(t-0.5)}(t > 0.5hr), \quad (2)$$

where C_{serum_van} is the vancomycin concentration in serum (unit: ug/mL), and t is time (unit: hour), the start of drug infusion is marked as $t = 0$.

Change of Cefepime Concentration in Serum With Time Following Infusion

Based on the measured cefepime concentration in blood after 2 g infusion in 30 minutes,[25] the serum level of cefepime after 2 g infusion in 30 minutes with a constant rate may be expressed as biexponential equation as follows:

$$c_{serum_cef} = \frac{163.0825}{0.5} \cdot t \quad (t \le 0.5\,hr), \quad (3)$$

$$c_{serum_cef} = 338.3 \cdot e^{-2.825\cdot t} + 99.07 \cdot e^{-0.4103\cdot t} \quad (t > 0.5\,hr), \quad (4)$$

where c_{serum_cef} is the cefepime concentration in serum (unit: ug/mL), and t is time (unit: h), the start of drug infusion is marked as $t = 0$.

Net Charge of Antibiotics as a Function of pH

A general expression derived from the Henderson–Hasselbalch equation, for the net charge (z) of a molecule is as follows:[26]

$$z = \sum_p (+1) \cdot (1 - \alpha) + \sum_n (-1) \cdot \alpha, \quad (5)$$

$$\alpha = \frac{1}{10^{pKa - pH} + 1}, \quad (6)$$

where P represents the functional groups like—NH_3^+, $=NH_2^+$, $\equiv NH^+$, and n represents the functional groups like—COOH, −SH, −PhOH, and pKa is an acidity constant. Vancomycin contains six functional groups, namely, −COOH, $-NH_2$, $-NHCH_3$, −OH, −OH, −OH. The pKa's related to each of these groups are: 2.18, 7.75, 8.89, 9.59, 10.4, 12.0.[27,28] For example, the net charge of vancomycin calculated with the expression above is 0.83 at pH = 7, and 0.65 at pH = 7.4. Cefepime contains three functional groups, namely, −COOH, $-NH^+$, −NH. The pKa's related to each of these groups are 1.12, 3.07, 10.8.[29]

For diskitis, recommended antibiotics dosage regimens are: vancomycin 15 mg/kg for every 12 hours, and cefepime 2 g every 12 hours.[17] In the present study, the minimal inhibitory concentration (MIC) for vancomycin was set to be 1 ug/mL.[30] The MIC for cefepime was set to be 8 ug/mL.[31] In this study, the IV infusions of various dosages of vancomycin and cefepime were simulated. The antibiotic concentrations in nondegenerated and moderately degenerated discs were investigated.

RESULTS

Intravenous Infusion of 1 g Vancomycin Together With 2 g Cefepime in 30 Minutes, Repeated Every 12 Hours

In the Nondegenerated Disc

Although the peak concentration of vancomycin in the serum could reach to the level around 64.0 times of its MIC (*i.e.*, 1 ug/mL), the vancomycin concentration in the disc was predicated to fluctuate between 17.0 and 31.0 times of its MIC in about 2 days (Figure 2A). It was found that, starting from the 3rd cycle (*e.g.*, 36th h point), the pattern for temporal change in vancomycin concentration in the disc is repeatable, see Figure 2A.

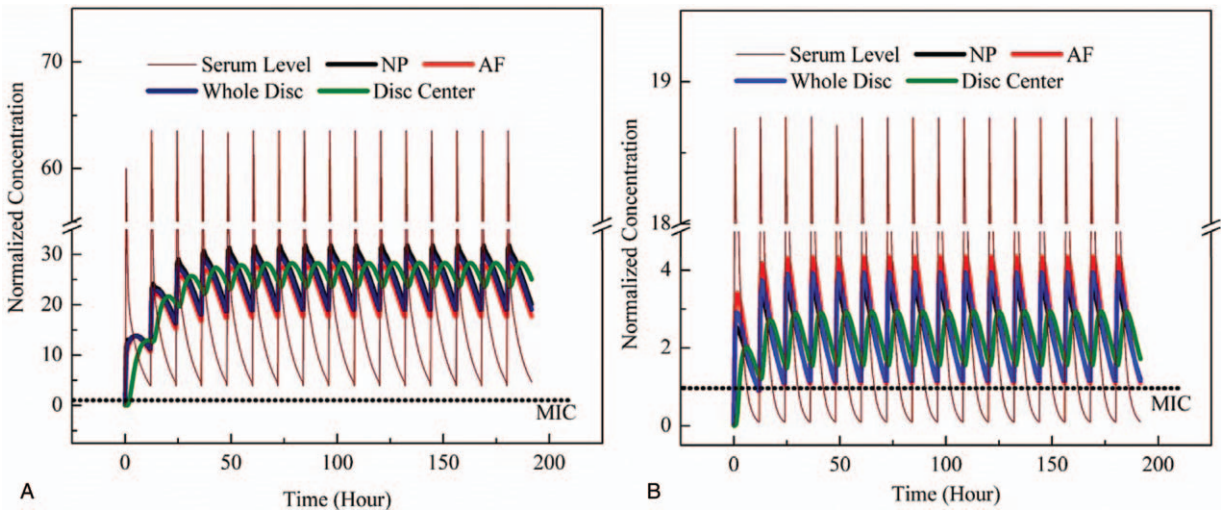

Figure 2. A, Vancomycin concentration in nondegenerated disc (normalized to its MIC, *i.e.*, 1 ug/mL). Vancomycin (1 g) was infused intravenously in 30 minutes every 12 hours. **B**, Cefepime concentration in nondegenerated disc (normalized to its MIC, *i.e.*, 8 ug/mL). Cefepime (2 g) was infused in 30 minutes every 12 hours.

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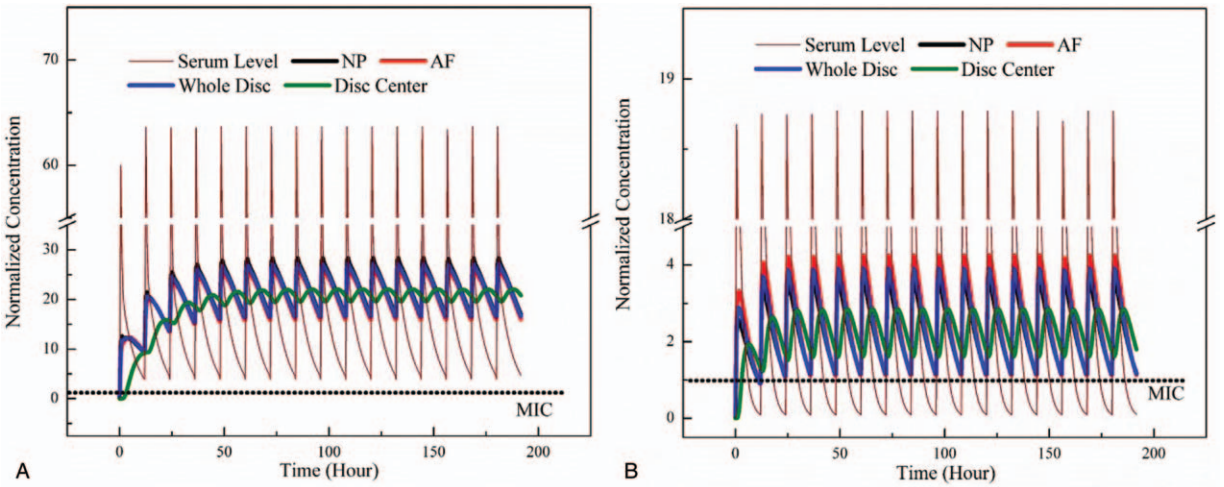

Figure 3. A, Vancomycin concentration in moderately degenerated disc (normalized to its MIC, *i.e.*, 1 ug/mL). Vancomycin (1 g) was infused intravenously in 30 minutes every 12 hours. **B**, Cefepime concentration in moderately degenerated disc (normalized to its MIC, *i.e.*, 8 ug/mL). Cefepime (2 g) was infused in 30 minutes every 12 hours.

For cefepime, while its level in the serum could be as high as 18.7 times of its MIC (*i.e.*, 8 ug/mL), the level in the disc was predicted to fluctuate between 1.1 and 4.2 times of its MIC in about 2 days (Figure 2B). Similar to vancomycin, the pattern of temporal change in cefepime concentration in the disc is repeatable starting from the 3rd cycle (*e.g.*, 36th h point), see Figure 2B.

In the Moderately Degenerated Disc

Vancomycin concentration in the disc was predicated to fluctuate between 16.8 and 28.4 times of its MIC in about 2 days (Figure 3A). Starting from the 3rd cycle (*e.g.*, 36th h point), the pattern for temporal change in vancomycin concentration in the disc is repeatable, see Figure 3A.

For cefepime, its concentration in the disc was predicated to fluctuate between 1.2 and 4.2 times of its MIC in about 2 days (Figure 3B). Similar to vancomycin, the pattern of temporal change in cefepime concentration in the disc is repeatable starting from the 3rd cycle (*e.g.*, 36 h point), see Figure 3B.

It was predicted that moderate disc degeneration had a considerable influence (up to 25%) on the vancomycin concentration in the disc, see Figure 4A. The vancomycin concentration in the moderately degenerated disc was much smaller than that in the nondegenerated disc, whereas the moderate degeneration had less effect on the transport of cefepime in the disc (Figure 4B).

Intravenous Infusion of 0.5 g Vancomycin Together With 1 g Cefepime in 30 Minutes Every 12 Hours

We also analyzed the case with half of the dose of each antibiotic, that is, 0.5 g of vancomycin and 1 g of cefepime were infused intravenously in 30 minutes every 12 hours, in the nondegenerated disc. It was predicted that vancomycin

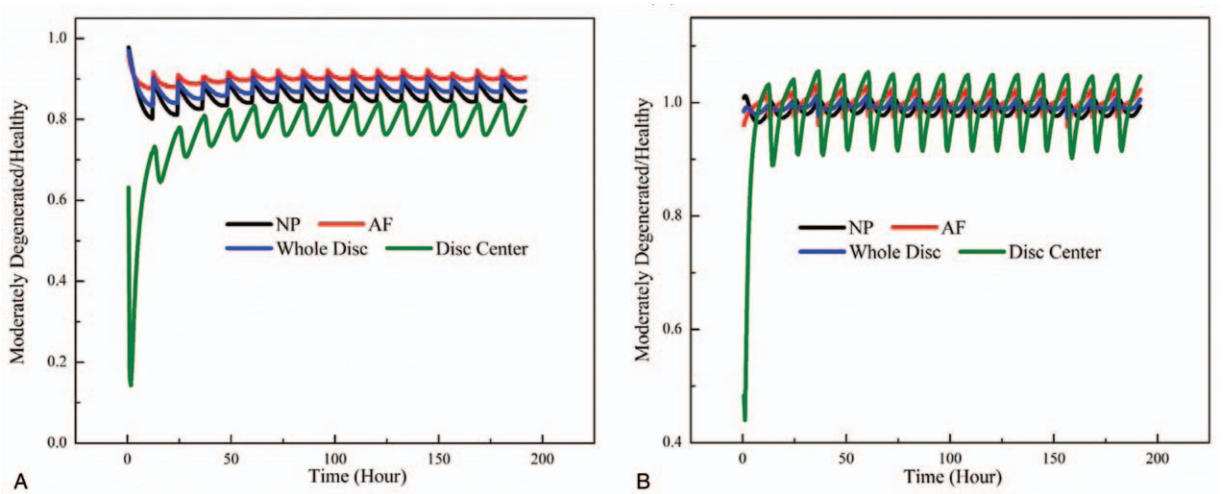

Figure 4. A, Comparison of vancomycin concentration in moderately degenerated disc to that in nondegenerated disc. Vancomycin (1 g) was infused intravenously in 30 minutes every 12 hours in both discs. **B**, Comparison of cefepime concentration in moderately degenerated disc to that in nondegenerated disc. Cefepime (2 g) was infused in 30 minutes every 12 hours in both discs.

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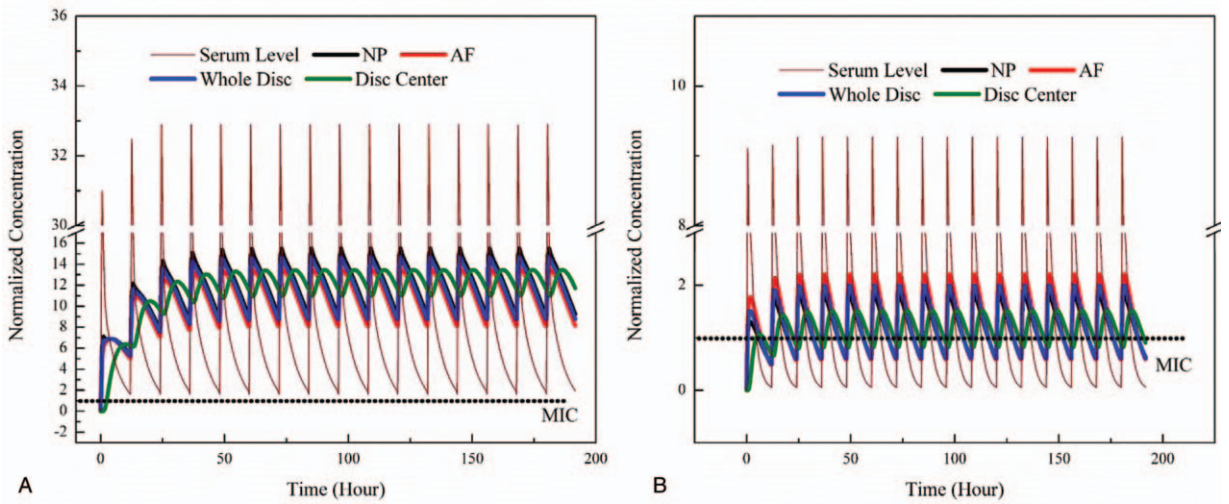

Figure 5. A, Vancomycin concentration in nondegenerated disc (normalized to its MIC, *i.e.*, 1 ug/mL). Vancomycin (0.5 g) was infused intravenously in 30 minutes every 12 hours. **B**, Cefepime concentration in nondegenerated disc (normalized to its MIC, *i.e.*, 8 ug/mL). Cefepime (1 g) was infused in 30 minutes every 12 hours.

concentration in each of the disc region was above its MIC level after around 30 minutes (Figure 5A), whereas cefepime concentration dropped below its MIC level from time to time (Figure 5B).

DISCUSSION

In this study, we quantitatively evaluated the vancomycin and cefepime penetration into nondegenerated and moderately degenerated discs using a finite element method. We analyzed the concentration distributions of vancomycin and cefepime in different regions of the discs and studied the difference in drug concentration between nondegenerated and degenerated discs.

Our results indicate that vancomycin is accumulated in the disc much easier than cefepime. The reason behind such phenomenon is that vancomycin is positively charged, while the cefepime is close to neutral, in both the nondegenerated disc and moderately degenerated disc simulated in our model. This is because electric charge of an antibiotic has great effects on kinetics of its penetration into the disc, with higher concentrations and uptakes for positively charged antibiotics than those for the noncharged and negatively charged ones.[7–13]

The reason that disc degeneration has considerable influence (up to 25%) on vancomycin concentration but less on cefepime in the disc is also due to the charge effects. The positively charged vancomycin is influenced by the change in fixed charge density in the degenerated disc (relative to that in the nondegenerated disc), while the electrically neutral cefepime is not influenced by the change of the charge environment in degenerated disc (relative to that in the nondegenerated disc).

In this study, the diffusion coefficient of vancomycin in water at 37° C, 2.83×10^{-10} m^2/s,[21] was used to simulate vancomycin diffusion to disc without binding effect. The value of diffusion coefficient in the disc was estimated by the diffusion model:[32] $\frac{D}{D_0} = \exp\left[-\alpha\left(\frac{r}{\sqrt{\kappa}}\right)^{\beta}\right]$, where D is the diffusivity of vancomycin in disc, r is the Stokes radius of vancomycin, κ is the Darcy permeability of the disc, D_0 is vancomycin diffusivity in water, α and β are two positive parameters whose values depend on the disc material and structure. The average value of vancomycin diffusivity in the AF region of the nondegenerated disc in our study at 37° C was around: 10.6×10^{-11} m^2/s. A recent study on the apparent diffusivity of vancomycin in the AF of bovine coccygeal under 20% compressive strain at room temperature (~23° C) was reported to be around $7.94\pm2.00\times10^{-12}$ m^2/s, suggesting there could be a reversible binding effect between vancomycin and disc tissue.[33] This binding effect was simulated by using a 10 times (×10) smaller value of vancomycin diffusion coefficient in our study. It was found that a smaller value of diffusion coefficient slowed down vancomycin penetration into disc, but did not affect the concentration distribution of vancomycin at steady state significantly, see Figure 6. For the case with ×10 smaller diffusion coefficient, the concentration of vancomycin in the disc center increased monotonically (no oscillation) with time (Figure 6, lower-right figure). This is because the characteristic diffusion time at disc center (*e.g.*, Time = Length^2/Diffusivity) is much longer than the infusion interval (*e.g.*, 12 h).

It is recommended that the safe infusion rate of vancomycin (no side effects) is <500 mg/30 min in adults with normal renal functions.[22] Infusion rate higher than this value may lead to high transient peak concentrations that could cause significant side effects such as tachycardia, pruritus at the site of injection, general flushing, nausea, palpitations, substernal pressure, lightheadedness.[22] Our results of vancomycin infusion of 500 mg in 30 minutes every 12 hours showed that this regimen is effective for maintaining vancomycin concentration above the MIC level in the disc. This indicates that the clinically recommended

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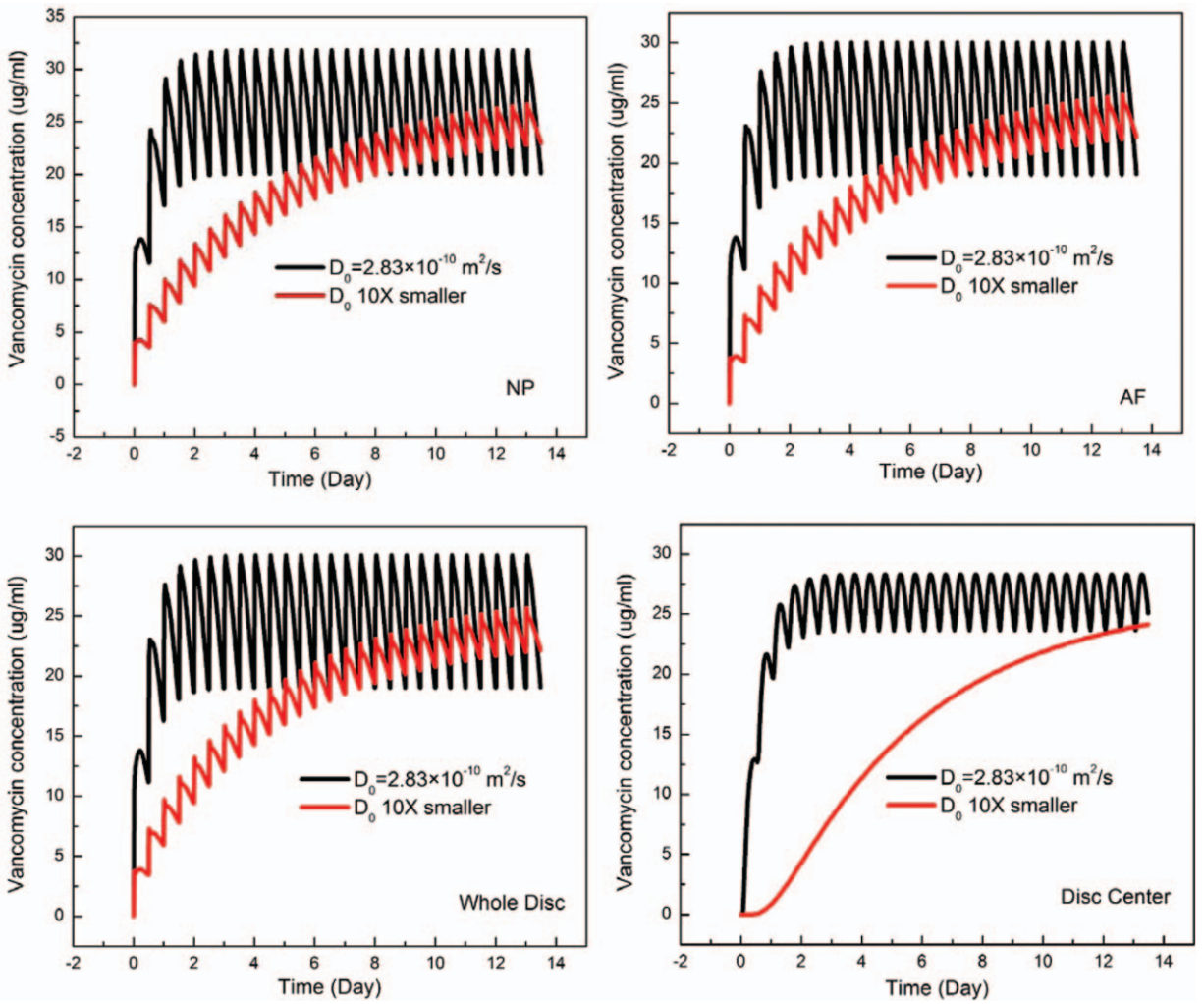

Figure 6. Comparison of Vancomycin concentration in nondegenerated disc (unit: ug/mL) between diffusivity (in water at 37° C) = 2.83×10^{-10} m^2/s and diffusivity 10 times smaller (to simulate the case with a binding effect). Vancomycin (1 g) was infused intravenously in 30 minutes every 12 hours. NP indicates nucleus pulposus; AF, annulus fibrosus.

dose of 1 g vancomycin could be reduced by half, with the same treatment effects without causing potential side effects.

It is recommended that cefepime concentration at 4.3 times of its MIC is optimal for bacteria (*e.g.*, gram-negative infections) killing.[34] Our results showed that the concentration of cefepime in the disc is around one to four times of its MIC with the regimen of 2 g in 30 minutes every 12 hours (Figure 2B), indicating that the currently used dose of cefepime may be not high enough for optimally treating infected discs.

In this study, the effects of daily mechanical loading on vancomycin and cefepime penetration have not been considered. Studies in the literature have shown that dynamic loading may increase neutral solute transport and concentration in the disc.[35–40] The binding effect of antibiotic transport in the disc was not considered specifically, and its effect was taken into consideration by using lower value of diffusion coefficient, due to the very limited information on binding kinetics of antibiotics in the literature. As has been discussed and shown in Figure 6, this binding effect may slow down vancomycin penetration into the disc, but may not significantly affect the concentration distribution of vancomycin at steady state. Another limitation of this study is that only a disc with regular geometry and size was simulated. The future studies should consider patient-specific disc geometry as well as the anisotropy of the annulus fibrosus. In consideration with these limitations of the current model, caution should be used when choosing the dosage for treating specific patient based on our recommendations for dosage reported in this study. Note that the transport of antibiotic drugs in the human body is more complex, especially in patients with disc infection. However, this research provides a basis for using antibiotic drugs more accurately in the future.

In summary, the temporal and spatial distributions of vancomycin and cefepime, two commonly used antibiotics in clinic for treating disc infection related diseases, were quantitatively studied with a three-dimensional finite element model developed based on the mechano-electrochemical multiphasic mixture theory. Knowledge of the quantitative distribution of antibiotics in the human discs

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is very important for providing quantitative guidance on selecting proper dose and dosing intervals on a patient-specific basis.

➢ Key Points

- ❑ Transport of two antibiotics (vancomycin and cefepime) in the human intervertebral disc administrated through intravenous infusion with clinically recommended dose and dosage [*e.g.*, vancomycin (1 g every 12 h) and cefepime (2 g every 12 h)] was simulated with a computational model.
- ❑ The temporal and spatial distributions of vancomycin and cefepime in the disc were quantitatively investigated.
- ❑ It was found that vancomycin concentration in the disc fluctuated between 17.0 and 31.0 times of its minimum inhibitory concentration (MIC, 1 ug/mL) for the case of 1 g infusion in 30 minutes, repeated every 12 hours; and cefepime concentration in the disc fluctuated between 1.1 and 4.2 times of its MIC (*i.e.*, 8 ug/mL) for the case of 2 g infusion in 30 minutes, repeated every 12 hours.
- ❑ Moderate disc degeneration had a considerable influence on the vancomycin concentration in the disc, compared with that on cefepime concentration.

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